

Forecasting Energy Load

Short-term forecasting using classical
machine learning and deep learning
models on German energy data



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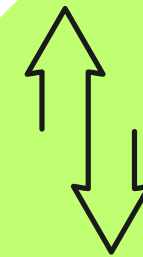
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Why is electricity demand forecasting important?



Electricity demand changes every 15 minutes and must be balanced in real time



Inaccurate forecasts lead to higher operational costs



Reliable forecasts help ensure stability of the power grid



Renewable energy (wind & solar) increases uncertainty



Data & Forecasting Objective



Goal: Learn short-term patterns

Use last **7** days of data
(**672** observations)



Predict next **24** hours
(**96** observations)



Wind, Solar



Coal, Gas, Oil



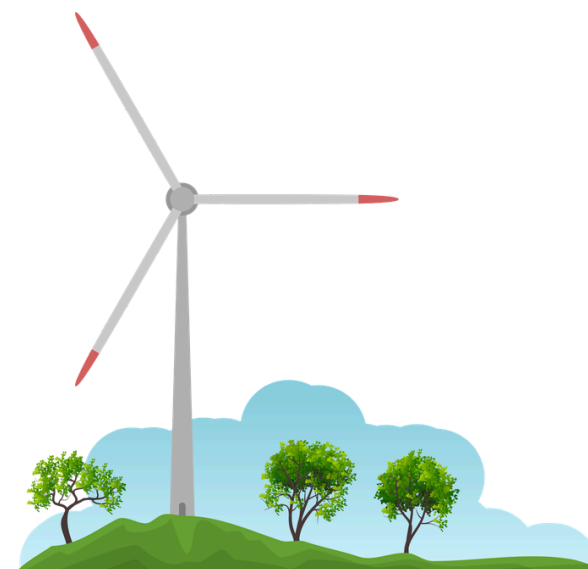
Reservoir, Pumped storage



Other sources
and time features



Load (incl. self-consumption)



Modeling Approaches



Tree based Models

- 1 XGBoost
- 2 LightGBM
- 3 CatBoost



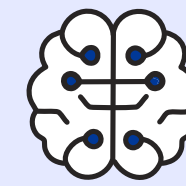
Many external features



Sliding window
We use sliding window approach for training



Input 7 days
Past 7 days of data (multiple features)



Neural Networks

- 1 RNN
- 2 GRU
- 3 LSTM

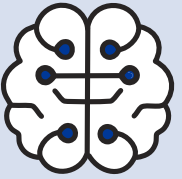


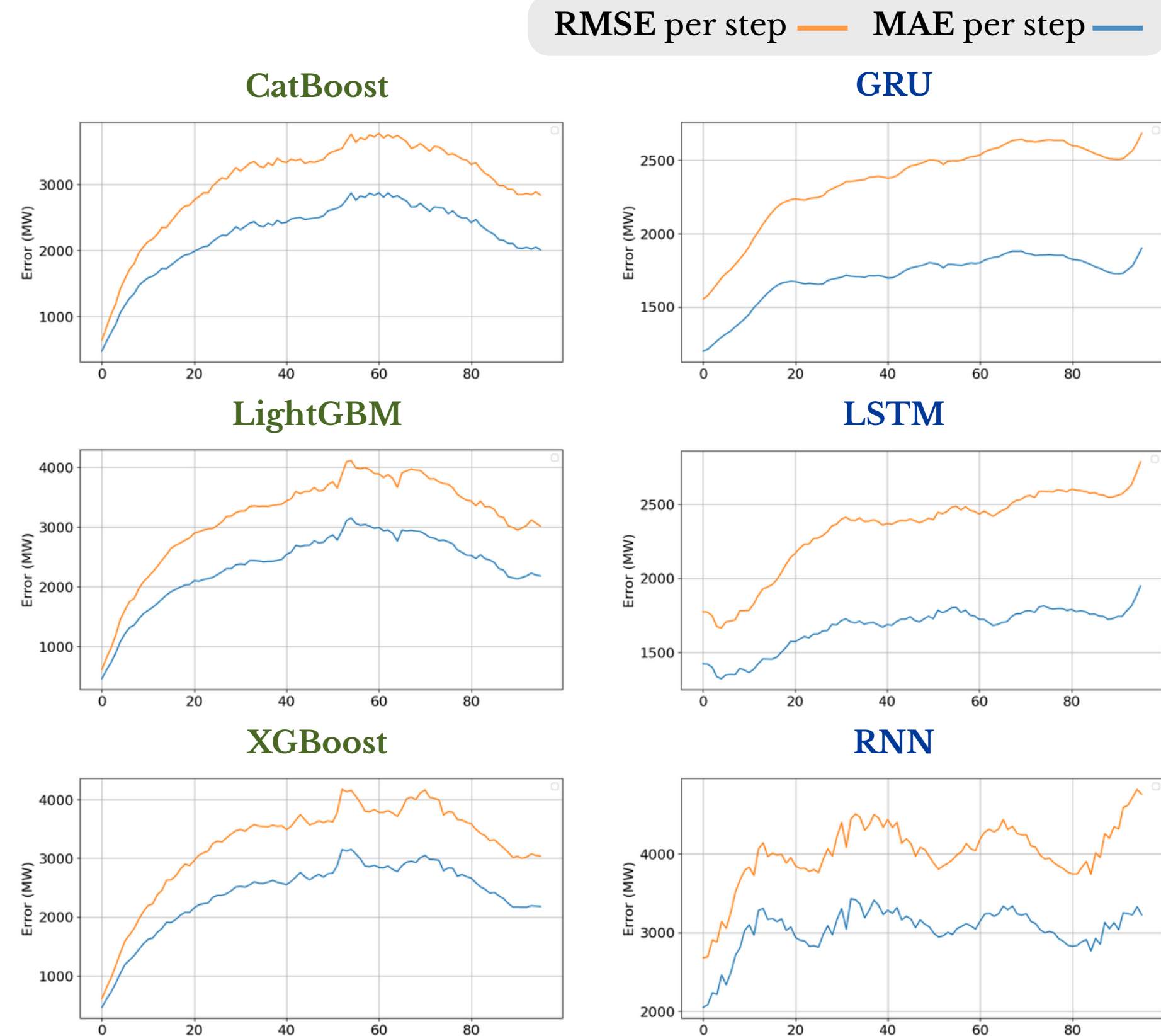
Only time features



Goal 24 hours
Forecast the next 24 hours (96 steps)

Model Performance Results

	Model	R2	RMSE	MAE
 Neural Networks	GRU	0,94	2349	1666
	LSTM	0,939	2382	1704
	RNN	0,825	4031	3035
 Tree based Models	CatBoost	0,885	2211	1644
	LightGBM	0,879	2266	1702
	XGBoost	0,869	2361	1778



Key Takeaways

- CatBoost achieves the best overall performance with the lowest error and highest R2 among **tree models**
- LSTM and GRU performs competitively with the tree-based models.
- Deep learning models explain data variance better.

Interpretation & Limitations



Interpretation

- Tree based models achieve lower prediction errors (RMSE, MAE)
- Neural Networks show higher R2 but larger errors
- Prediction errors increase with forecast horizon
- Short-term electricity demand is highly predictable

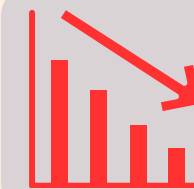


Tree models are more accurate in current circumstances, while Neural Network better capture overall data structure.



Limitations

- ! **No** external data such as weather or holidays included
- ! Separated models for each of 96 horizons (**for tree-based models**)
- ! Limited hyperparameter tuning
- ! Deep learning models trained only on data since 2022 due to memory constraints



Reduced training data may limit the performance of deep learning models.



Future idea - incorporating external features (e.g. weather, calendar effects) and using more data could improve performance of deep learning



Thank you for your attention!

Questions?



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